Multi-Model LLM Routing Using Federated Proximal Policy Optimization Under Quality and Resource Constraints

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***Abstract*—The increasing diversity of Large Language Models (LLMs) necessitates intelligent routing mechanisms that balance response quality, inference cost, and latency across heterogeneous workloads. Existing multi-LLM routing approaches largely rely on centralized training or static policies, limiting adaptability when query and model-performance data are distributed across environments. This paper proposes a Federated Proximal Policy Optimization (Federated PPO) framework for adaptive multi-LLM routing, where distributed routing agents independently learn model-selection policies from local query and performance feedback without sharing raw data. LLM selection is formulated as a constrained contextual decision problem, optimized using PPO to learn routing policies over heterogeneous candidate models. Local policy updates are periodically aggregated to construct a global routing policy, enabling cross-environment knowledge sharing while preserving data locality. The framework is evaluated across question answering, summarization, code generation, translation, and sentiment analysis using multiple LLMs. Performance is compared with single-model, conventional multi-LLM, and non-federated routing strategies using quality, cost, latency, and resource-efficiency metrics. Results demonstrate the potential of Federated PPO-based routing to provide adaptive, efficient, and privacy-aware model selection for scalable heterogeneous LLM serving.**

***Index Terms*—Large Language Models, multi-LLM routing, Federated Learning, Proximal Policy Optimization, reinforcement learning, model selection, resource-aware routing.**

1. Introduction

The rapid adoption of Large Language Models (LLMs) has enabled applications across diverse tasks, including question answering, summarization, code generation, translation, and sentiment analysis. However, no single LLM is consistently optimal across workloads, as models differ in response quality, inference cost, latency, and resource requirements. Intelligent multi-LLM routing is therefore required to select an appropriate model for each request while balancing these objectives.

Reinforcement learning (RL), particularly Proximal Policy Optimization (PPO), provides a framework for learning adaptive routing policies. In this work, each request is independently mapped to a routing action, making the problem a contextual decision problem optimized using PPO. The routing reward jointly considers response quality, inference cost, latency, and a Resource Load Proxy (RLP), while quality and resource constraints are incorporated into the routing objective.

Centralized training becomes challenging when routing data originate from heterogeneous clients with task-specific and non-IID workloads. Centralizing such data may introduce privacy and data-sharing constraints. Federated Learning (FL) addresses this challenge by enabling local training while aggregating model parameters without exchanging raw client data.

This work proposes a **Federated PPO with Federated Averaging (FedAvg)** framework for multi-LLM routing across heterogeneous task-specific clients. Five clients represent question answering, summarization, code generation, translation, and sentiment analysis. Each client performs

local PPO training, and the resulting parameters are aggregated using sample-size-weighted FedAvg to obtain a global routing policy.

The framework is evaluated across 5,000 requests using three candidate LLMs—GPT-OSS-120B, GPT-OSS-20B, and Qwen 3.6-27B. Federated PPO is compared with centralized PPO, local PPO, Logistic Regression, LSTM, and fixed-model routing baselines using constraint satisfaction, quality, cost, latency, and Resource Load Proxy. On the held-out 727-request test set, Federated PPO + FedAvg achieves 79.09% constraint satisfaction and 81.98% quality satisfaction.

The main contributions are:

1) **Multi-objective LLM routing:** A PPO-based routing formulation that jointly considers quality, cost, latency, and resource demand.
2) **Federated PPO routing:** A federated framework enabling heterogeneous task-specific clients to collaboratively learn a global routing policy using FedAvg without sharing raw data.
3) **Constraint-aware optimization:** Quality and resource constraints are incorporated into the routing objective to balance service quality and operational efficiency.
4) **Comprehensive evaluation:** The proposed approach is evaluated against centralized, local, learned, and fixed-model routing baselines on a held-out test set.

The remainder of this paper is organized as follows. Section II presents related work. Section III describes the proposed methodology. Section IV presents the experimental setup. Section V discusses the results, and Section VI concludes the paper.

2. Related Work

2.1. RL-Based LLM Routing

The increasing use of Large Language Models (LLMs) creates a trade-off between model capability and inference cost, latency, memory, and energy consumption. Model-selection and routing methods address this trade-off by selecting an appropriate model for each request rather than invoking the same model for all inputs. Ghosh *et al.* investigate adaptive contextual-bandit model selection [1], while Xia *et al.* propose TI-UCB for online LLM selection [2]. Recent LLM routing systems further incorporate cost, latency, quality-of-service, and resource constraints [3], [4], [5], [6].

Reinforcement learning provides another approach by learning routing policies from interaction outcomes. HierRouter formulates multi-LLM coordination as a finite-horizon decision process and applies PPO for specialized-model selection [7]. Router-R1, xRouter, CORL, and SeqRoute further investigate RL-based routing and orchestration under performance, budget, and resource considerations [8], [9], [10], [11]. These studies demonstrate that learned policies can adapt model selection to request characteristics and system objectives. However, the routing policy is predominantly trained in a centralized setting, where the training information from different tasks or environments is available to a common learner.

2.2. Federated Learning for LLMs and Federated RL

Federated Learning (FL) enables multiple clients to collaboratively train models without centralizing their raw local data. Federated LLM studies identify privacy, non-IID data, communication overhead, and client heterogeneity as key challenges [12], [13], [14]. OpenFedLLM demonstrates federated LLM training over decentralized private data [15], while FlexLoRA and FedPSF investigate heterogeneous client resources, non-IID data, and fairness considerations [16], [17]. These approaches primarily address collaborative fine-tuning or adaptation of shared LLMs rather than inference-time selection among independent heterogeneous LLM services.

Federated reinforcement learning extends this paradigm to decentralized policy optimization. Zhou *et al.* study federated offline RL [18], while federated PPO approaches demonstrate collaborative policy optimization across heterogeneous environments and resource-constrained settings [19], [20], [21], [22], [23]. These studies establish the feasibility of combining local policy optimization with federated aggregation, but their applications primarily concern domain-specific control problems rather than LLM model routing.

2.3. Federated LLM Routing and Research Gap

The most directly related work is "Federate the Router" by Askin *et al.*, which applies federated learning to LLM routing using MLP and K-means routers trained from decentralized query–model evaluations [24]. The approach improves the accuracy–cost trade-off while preserving query privacy. However, it focuses primarily on the accuracy–cost frontier and does not learn a federated policy that jointly incorporates response quality, monetary cost, latency, and resource load.

Table 1 summarizes the positioning of representative prior approaches. Existing work addresses individual aspects of efficient LLM routing, reinforcement-learning-based routing, federated LLM learning, or federated policy optimization. The combination of federated policy learning, heterogeneous task-specific clients, multi-objective routing, and explicit quality and resource constraints remains comparatively underexplored.

Here, "Qual.+Cost" denotes explicit joint consideration of response quality and monetary cost, while "+Lat./Resource" denotes the inclusion of latency and/or resource-related objectives. "Task-Het." indicates evaluation or learning across heterogeneous task-specific clients.

Note: "Resource" refers to the resource-related objective reported by each prior work and does not imply use of the LLM-specific Resource Load Proxy defined in this work.

To address this gap, this work proposes a **Federated PPO with FedAvg** framework for multi-LLM routing. PPO policies are trained locally on heterogeneous task-specific clients and collaboratively aggregated using sample-size-weighted FedAvg. The routing objective jointly considers response quality, inference cost, latency, and resource load, with explicit quality and resource constraints. The resulting global policy enables adaptive per-request model selection while keeping raw client data local.

3. Proposed Federated PPO Routing Method

This work formulates per-request multi-LLM selection as a federated reinforcement learning problem in which heterogeneous task-specific clients collaboratively learn a routing policy without exchanging raw local data. Let the candidate model set be $\mathcal{M} = \{M_1, M_2, M_3\}$. For each request, the router observes request and candidate-model characteristics and selects one model. Since requests are evaluated independently without state carryover, the problem can be viewed as a constrained contextual bandit optimized using PPO.

3.1. Routing State and Action Space

The routing state contains 14 features: two request-level features (complexity score and input-token count) and four normalized model-level metrics—quality, cost, latency, and Resource Load Proxy (RLP)—for each candidate LLM. Thus,

$$s_i = [x_i, \phi(M_1), \phi(M_2), \phi(M_3)], \tag{1}$$

where $x_i = [\text{complexity}_i, \text{inputTokens}_i]$ and $\phi(M_j) = [Q_j, C_j, L_j, R_j]$.

The RLP is defined as

$$R_s = (T_{\text{in}} + T_{\text{out}}) \times L, \tag{2}$$

where T_{in}, T_{out}, and L denote input tokens, output tokens, and inference latency, respectively. The discrete action space is $\mathcal{A} = \{0, 1, 2\}$, corresponding to M_1, M_2, and M_3, and the PPO policy $\pi_\theta(a|s)$ produces a probability distribution over these models.

TABLE 1. COMPARISON OF REPRESENTATIVE APPROACHES WITH THE PROPOSED METHOD

Approach	RL	Fed.	Qual.+Cost	+Lat./Resource	Task-Het.
Contextual selection [1]	–	–	–	–	–
TI-UCB [2]	–	–	–	–	–
HierRouter / Router-R1 / xRouter [7], [8], [9]	Yes	–	Yes	–	–
CORL [10]	Yes	–	Yes	–	–
SeqRoute [11]	Yes	–	Yes	–	–
OpenFedLLM [15]	–	Yes	–	–	Yes
FlexLoRA [16]	–	Yes	–	–	Yes
Federated PPO [21]	Yes	Yes	Partial	Partial	Yes
Federate the Router [24]	–	Yes	Yes	–	Yes
Proposed	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

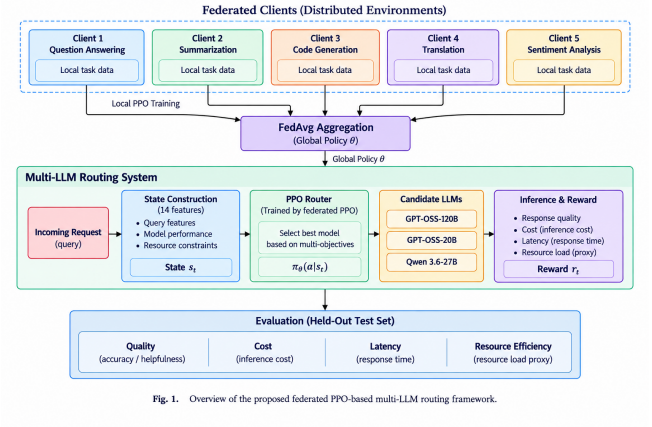


Fig. 1. Overview of the proposed federated PPO-based multi-LLM routing framework.

Figure 1. Overview of the proposed federated PPO-based multi-LLM routing framework.

3.2. Multi-Objective Routing Reward

The routing policy jointly optimizes response quality and inference efficiency. The normalized reward is

$$R = 0.40Q_{\text{norm}} - 0.20C_{\text{norm}} - 0.20L_{\text{norm}} - 0.20S_{\text{norm}} - P_Q - P_R, \tag{3}$$

where Q, C, L, and S denote quality, monetary cost, latency, and RLP, respectively. Quality receives the highest weight, while the three efficiency terms receive equal weights.

Quality and resource constraints are imposed through

$$P_Q = \begin{cases} 1, & Q < Q_{\min}, \\ 0, & Q \geq Q_{\min}, \end{cases} \qquad P_R = \begin{cases} 1, & R_s > R_{\max}, \\ 0, & R_s \leq R_{\max}. \end{cases} \tag{4}$$

Here, $Q_{\min}$ is the median of the training-quality distribution and $R_{\max}$ is the 75th percentile of the training RLP distribution. Both thresholds are derived exclusively from training data to prevent test-set information leakage.

The quality metric is task dependent: token-overlap F1 for question answering, binary correctness for sentiment analysis, ROUGE-L F1 for summarization and code generation, and sentence-level BLEU-4 with equal n-gram weights and Method-1 smoothing for translation. ROUGE-L is used for code generation as a lightweight, scalable reference-based signal without requiring sandboxed execution. However, it measures lexical and sequence overlap rather than functional correctness and may penalize valid alternative implementations or reward textually similar but incorrect code. Execution-based metrics such as pass@k are therefore identified as future complementary or replacement measures.

3.3. Federated PPO Learning

Five heterogeneous clients represent question answering, summarization, code generation, translation, and sentiment analysis. Each client maintains task-specific non-IID data and performs local PPO training. Let $\theta^{(r)}$ denote the global policy at round r. It is distributed to each client as

$$\theta_k^{(r,0)} = \theta^{(r)}. \tag{5}$$

After local PPO optimization, client k obtains $\theta_k^{(r+1)}$. The policies are aggregated using sample-size-weighted FedAvg:

$$w_k = \frac{n_k}{\sum_{j=1}^{K} n_j}, \tag{6}$$

$$\theta^{(r+1)} = \sum_{k=1}^{K} w_k \theta_k^{(r+1)}. \tag{7}$$

The aggregation operates on the trainable policy and value-network parameters; raw prompts, datasets, gradients, trajectories, and optimizer states remain local.

3.4. Federated Training Procedure

The training procedure runs for 20 federated rounds, with all five clients performing 500 local PPO timesteps per round. Algorithm 1 summarizes the procedure.

Thus, the aggregate local training budget is

$$20 \times 5 \times 500 = 50{,}000 \tag{8}$$

timesteps across all client-round combinations. The final global policy is then used to route the held-out test requests to one of the three candidate LLMs.

Algorithm 1 Federated PPO with FedAvg

Require: Initial policy $\theta^{(0)}$, client datasets $\{D_1, \ldots, D_K\}$, rounds R, local budget T

for $r = 1, \ldots, R$ **do**
  for each client $k = 1, \ldots, K$ in parallel **do**
    $\theta_k \leftarrow \theta^{(r-1)}$
    $\theta_k \leftarrow \mathrm{LocalPPO}(\theta_k, D_k, T)$
  end for
  $\theta^{(r)} \leftarrow \sum_{k=1}^{K} w_k \theta_k$
end for
return $\theta^{(R)}$

4. Experimental Setup

4.1. Dataset and Task Distribution

The experiments were conducted on a dataset containing 5,000 prompts distributed uniformly across five task categories: Question Answering (QA), Summarization, Code Generation, Translation, and Sentiment Analysis. Each task contains 1,000 prompts. The dataset was divided at the request level using a 70%/15%/15% split for training, validation, and testing, respectively, resulting in 3,500 training prompts, 750 validation prompts, and 750 test prompts. Incomplete benchmark requests were excluded from the final evaluation, resulting in 727 complete held-out test requests. To represent heterogeneous federated clients, each task category was assigned to a separate client, resulting in five clients with task-specific data distributions.

4.2. Candidate LLMs

Three LLMs were considered as candidate models for routing: M1 (`openai/gpt-oss-120b`), M2 (`openai/gpt-oss-20b`), and M3 (`qwen/qwen3.6-27b`). All models were accessed through the Groq API. Task-specific system prompts were used for each request, with the inference temperature fixed at zero. For every model invocation, input and output tokens, response latency, and API cost were recorded for evaluating routing performance.

4.3. Baselines

The proposed Federated PPO with FedAvg was compared with seven routing baselines. Fixed M1, Fixed M2, and Fixed M3 always select the corresponding LLM for every request. Logistic Regression performs supervised classification-based routing, while LSTM provides a neural sequence-based routing approach. Centralized PPO trains a single PPO policy using centralized training data, whereas Local PPO independently trains a PPO policy at each client without federated aggregation. These baselines provide comparisons across fixed-model, supervised-learning, centralized reinforcement-learning, and local reinforcement-learning approaches.

4.4. Evaluation Metrics

Routing performance was evaluated using constraint satisfaction, quality satisfaction, resource satisfaction, average quality, average cost, average latency, and average resource load. Constraint satisfaction measures the percentage of requests satisfying both quality and resource constraints:

$$CS = \frac{N(Q \geq Q_{\min} \wedge R_s \leq R_{\max})}{N} \times 100. \tag{9}$$

Quality and resource satisfaction are defined as

$$QS = \frac{N(Q \geq Q_{\min})}{N} \times 100, \qquad RS = \frac{N(R_s \leq R_{\max})}{N} \times 100. \tag{10}$$

The resource-load proxy is calculated as

$$R_s = (T_{\text{in}} + T_{\text{out}}) \times L, \tag{11}$$

TABLE 2. PPO AND FEDERATED TRAINING CONFIGURATION

Parameter	Value
Policy architecture	MlpPolicy
Learning rate	0.0003
Rollout length (n_{steps})	64
Batch size	32
Discount factor (γ)	0.99
GAE parameter (λ)	0.95
Clipping range	0.2
PPO epochs	10
Entropy coefficient	0.01
Value-function coefficient	0.5
Maximum gradient norm	0.5
Federated rounds	20
Local PPO timesteps/round	500
Participating clients/round	5
Random seed	42

where T_{in} and T_{out} denote input and output tokens, respectively, and L denotes response latency. Average quality, cost, latency, and resource load are additionally reported to characterize the quality–efficiency trade-off.

4.5. Experimental Configuration

The routing agents were implemented using Stable-Baselines3 PPO with the configuration summarized in Table 2. All five clients participated in every federated round, and the final global policy was evaluated on the held-out test set.

5. Results and Discussion

5.1. Overall Routing Performance

The routing methods were evaluated on the held-out test set containing 727 complete requests. Fig. 2 compares constraint satisfaction, quality satisfaction, and resource satisfaction across the evaluated routing approaches. The results show that the PPO-based methods provide stronger quality–constraint performance than the supervised and fixed-model approaches. Centralized PPO achieves the highest constraint satisfaction of 79.64% and quality satisfaction of 82.94%, while the proposed Federated PPO achieves 79.09% and 81.98%, respectively. Thus, Federated PPO approaches the performance of centralized training while learning through federated aggregation.

5.2. Overall Test Results

Table 3 presents the complete test-set results. Federated PPO achieves a constraint satisfaction of 79.09% and a quality satisfaction of 81.98%, substantially outperforming Logistic Regression (70.56% and 74.14%) and

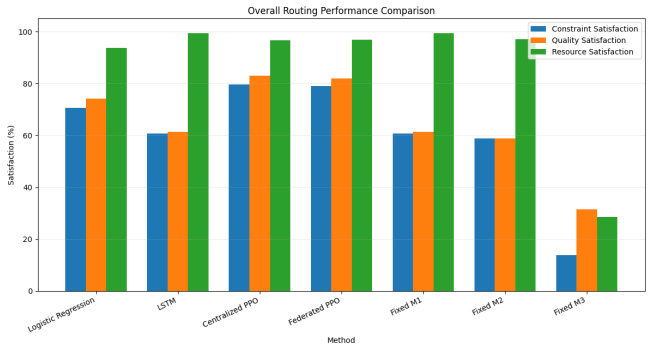


Figure 2. Overall routing performance on the held-out test set.

TABLE 3. OVERALL TEST-SET PERFORMANCE

Method	CS (%)	QS (%)	RS (%)	AQ	Cost (USD)	Latency (s)	Resource
Logistic Reg.	70.56	74.14	93.81	0.3027	0.000380	1.580	695.42
LSTM	59.70	60.11	96.70	0.1865	0.000127	0.798	452.47
Centralized PPO	**79.64**	**82.94**	96.56	**0.3710**	0.000302	1.255	501.93
Local PPO	76.62	79.92	96.56	0.3648	0.000297	1.224	464.67
Federated PPO	79.09	81.98	**96.84**	0.3673	**0.000259**	**1.154**	**450.46**
Fixed M1	60.80	61.35	99.45	0.1780	0.000140	0.827	263.52
Fixed M2	58.73	58.87	97.11	0.1755	**0.000105**	**0.742**	418.86
Fixed M3	13.76	31.50	28.47	0.1497	0.003207	6.720	8542.39

TABLE 4. PER-TASK MODEL SELECTION DISTRIBUTION BY FEDERATED PPO

Task Type	**M1 (%)**	**M2 (%)**	**M3 (%)**	**Total Requests**
Code Generation	37.01	**62.99**	0.00	127
Question Answering	**50.67**	49.33	0.00	150
Sentiment Analysis	12.67	26.00	**61.33**	150
Summarization	40.00	**60.00**	0.00	150
Translation	41.33	**58.67**	0.00	150

LSTM (59.70% and 60.11%). Centralized PPO obtains the highest overall quality, with an average quality of 0.3710. Federated PPO obtains a comparable average quality of 0.3673 while achieving lower average cost, latency, and resource load.

5.3. Task-Dependent Model Selection Behavior

To further investigate the adaptability of the learned policy across heterogeneous workloads, we analyzed the per-task model selection distribution of the Federated PPO router. Table 4 presents the routing distribution across the 727 held-out test requests for each task category.

The results provide direct evidence of task-dependent model-selection behavior. The federated policy does not apply a uniform or static routing distribution across all queries; rather, it dynamically adapts its decisions to the specific characteristics and requirements of each workload. As shown in Table 4, M2 is the dominant model for Code Generation (62.99%), Summarization (60.00%), and Translation (58.67%). In contrast, M1 is preferred for Question Answering (50.67%), while M3 is overwhelmingly selected for Sentiment Analysis (61.33%).

This dynamic distribution significantly strengthens the premise that the federated policy successfully adapts to heterogeneous client environments. By independently learning to route requests based on task-specific quality and resource constraints, the global policy maximizes overall system efficiency without requiring manual, rule-based intervention.

5.4. Constraint Satisfaction and Efficiency

Federated PPO provides a favorable balance between routing quality and resource efficiency. Compared with Local PPO, Federated PPO improves constraint satisfaction from 76.62% to 79.09% and quality satisfaction from 79.92% to 81.98%. Compared with Centralized PPO, the federated policy reduces constraint satisfaction by only 0.55 percentage points and quality satisfaction by 0.96 percentage points, while reducing average cost from 0.000302 USD to 0.000259 USD, latency from 1.255 s to 1.154 s, and resource load from 501.93 to 450.46.

The fixed-model baselines further demonstrate the need for adaptive routing. Although Fixed M2 provides the lowest average cost and latency, its quality satisfaction and average quality are substantially lower than those of the PPO-based methods. Fixed M3 performs poorly across all major metrics, indicating that selecting a single candidate model cannot consistently satisfy the combined quality and resource requirements.

5.5. Federated PPO Convergence

Fig. 3 presents the empirical convergence behavior of Federated PPO over 20 communication rounds. Constraint satisfaction, average quality, and average reward remain relatively stable during the first six rounds and then improve rapidly between rounds 7 and 10. After approximately round 10, the three measures remain within a relatively narrow range, indicating stabilization of the learned global routing policy.

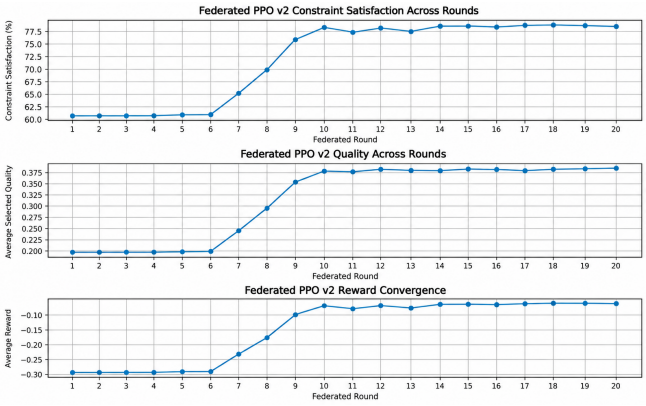


Figure 3. Federated PPO convergence over 20 communication rounds in terms of constraint satisfaction, average quality, and average reward.

TABLE 5. FIVE-SEED PPO RESULTS (MEAN ± STANDARD DEVIATION)

Method	CS (%)	QS (%)	AQ	Cost (USD)
Centralized PPO	79.42±0.21	82.48±0.42	0.3689±0.0030	0.000280±0.000016
Local PPO	78.21±1.96	81.54±1.98	0.3676±0.0053	0.000299±0.000008
Federated PPO	**79.12±0.12**	**82.01±0.37**	0.3673±0.0030	**0.000263±0.000013**

5.6. Federated, Centralized, and Local PPO

The robustness of the three PPO configurations was evaluated using five independent random seeds. Table 5 reports the mean and standard deviation across the five runs. Federated PPO achieves 79.12±0.12% constraint satisfaction and 82.01±0.37% quality satisfaction, which are close to Centralized PPO at 79.42±0.21% and 82.48±0.42%, respectively. In comparison with Local PPO, Federated PPO improves constraint satisfaction by 0.91 percentage points and quality satisfaction by 0.47 percentage points.

Federated PPO also provides the lowest average cost, latency, and resource load among the three PPO approaches. Its average cost is 0.000263 USD compared with 0.000280 USD for Centralized PPO and 0.000299 USD for Local PPO. Similarly, Federated PPO achieves an average latency of 1.1478 s and resource load of 432.70. Furthermore, the standard deviation of constraint satisfaction is substantially lower for Federated PPO (0.12%) than for Local PPO (1.96%), indicating more stable routing performance across random seeds.

6. Conclusion

Federated Proximal Policy Optimization (PPO) with Federated Averaging (FedAvg) provides an adaptive approach for multi-LLM routing across heterogeneous task-specific clients. The routing framework jointly considers response quality, inference cost, latency, and resource load, while enabling collaborative policy learning without exchanging raw client data.

Evaluation across 5,000 prompts covering question answering, summarization, code generation, translation, and sentiment analysis showed that Federated PPO provides competitive routing performance across heterogeneous workloads. On the held-out set of 727 complete requests, it achieved 79.09% constraint satisfaction, 81.98% quality satisfaction, and 96.84% resource satisfaction, with an average quality of 0.3673. It also achieved lower average cost, latency, and resource load than Centralized PPO and Local PPO in the reported test evaluation. Across five independent random seeds, Federated PPO achieved $79.12 \pm 0.12\%$ constraint satisfaction and $82.01 \pm 0.37\%$ quality satisfaction, with lower constraint- satisfaction variability than Local PPO.

The routing behavior further indicates adaptation to heterogeneous task requirements, with different candidate LLMs preferred across task categories. Federated training showed rapid improvement during the intermediate communication rounds followed by empirical stabilization after

approximately the tenth round. Overall, the results indicate that Federated PPO with FedAvg can achieve routing performance close to centralized PPO while maintaining favorable cost, latency, resource-efficiency, and robustness characteristics. Future work will consider larger and more diverse LLM pools, broader client heterogeneity, adaptive client participation, and evaluation on larger-scale real-world workloads.

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